

# A whole lot of reductions

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## 1 Preliminaries

**Definition 1.** A **tour** in a graph  $G = (V, E)$  is a sequence of vertices and edges adjacent to them in the graph that spans every vertex. We usually refer to the multiset of edges defined by a tour as  $T$ . The **cost** of a tour  $T$  is the sum of the weights of the edges in  $T$ , denoted as  $c(T) = \sum_{e \in T} c(e)$ .

This definition is equivalent to saying that a tour is a Eulerian multigraph that spans every vertex of  $G$ .

**Definition 2** (Metric TSP). Let  $G = (V, E)$  be a connected undirected graph. The **Metric Traveling Salesperson Problem** (Metric TSP) is defined as follows:

- **Instance:** A complete graph  $G = (V, E)$  which is connected by definition, and a cost function  $c : E \rightarrow \mathbb{R}_{\geq 0}$  that satisfies the triangle inequality:

$$c(u, v) \leq c(u, w) + c(w, v) \quad \text{for all } u, v, w \in V. \quad (1)$$

- **Objective:** Find a closed tour  $C$  in  $G$  such that the total weight  $c(C) = \sum_{e \in C} c(e)$  is minimized.

**Definition 3** (Quasi-Tour). A **quasi-tour**  $T_0$  in a graph  $G = (V, E)$  is a subset of edges in  $T$  removing edges from a finite set of closed tours. Each connected component of this multigraph is either a Eulerian multigraph or an isolated vertex.

A given multiset of edges  $T_0$  is a quasi-tour if and only if it is Eulerian. In the rest of this article, we are going to use a slightly modified version of this problem, enforcing some edges.

**Definition 4** ( $\text{TSP}_f$ ). An instance of the  **$\text{TSP}_f$**  problem is defined by a weighted metric graph  $G = (V, E_u \dot{\cup} E_f, c)$  and the objective is to find a tour that contains all the edges in  $E_f$  at least once and uses each edge at most twice.

We say that a tour  $T$  satisfying the conditions of  $\text{TSP}_f$  is a **valid tour**.

**Lemma 1.** *Let  $\varepsilon > 0$ . There is a polynomial time reduction mapping a given instance  $(G, c)$  of  $TSP_f$  to an instance  $(G', c')$  of Metric TSP such that, if  $T$  is the minimum weight spanning tour in  $G$  and  $T'$  is the minimum weight spanning tour in  $G'$ , then*

$$c(T) - \varepsilon \max_{e \in E_f} c(e) \leq c'(T') \leq c(T)$$

Then, we are not worsening the approximation factor of TSP by using  $TSP_f$  instead of TSP, as the optimal solution of  $TSP_f$  is at most  $\varepsilon \max_{e \in E_f} c(e)$  away from the optimal solution of TSP. For low enough values of  $\varepsilon$ , this difference is negligible, and thus the approximation factor is preserved.

## 2 Starting point

For the reduction, we will start from a common hard to approximate problem:

**Definition 5** (MAX-E3-LIN-2). *The **MAX-E3-LIN-2** problem is defined as follows:*

- **Instance:** *A set of  $n$  boolean variables  $x_1, x_2, \dots, x_n$  and a collection of  $m$  linear equations over  $\mathbb{F}_2$ , where each equation involves exactly three variables. Each equation is of the form:*

$$x_{i_1} + x_{i_2} + x_{i_3} = b_j, \tag{2}$$

*where  $b_j \in \{0, 1\}$  and  $i_1, i_2, i_3$  are distinct indices from  $\{1, 2, \dots, n\}$ .*

- **Objective:** *Find an assignment to the variables that maximizes the number of satisfied equations.*

**Theorem 1.** *For every  $\varepsilon \in (0, \frac{1}{4})$  and a sufficiently large integer  $k \geq k(\varepsilon)$ , the following partial decision subproblem  $Q(\varepsilon, k)$  of MAX-E3-LIN-2 is NP-hard: given an instance of MAX-E3-LIN-2 with  $m$  equations and exactly  $k$  occurrences of each variable, to decide if at least  $(1 + \varepsilon)m$  equations can be satisfied or if at most  $(\frac{1}{2} - \varepsilon)m$  equations can be satisfied by an optimal assignment.*

To prove some inapproximability results, it is more convenient for reductions if all equations have the same right hand side. The results obtained in such case can be enforced by flipping some occurrences of variables ( $\tilde{x} = 1 - x$ ). With that idea in mind, we can define a subproblem of  $Q(\varepsilon, k)$ .

**Definition 6.** *For  $b \in \{0, 1\}$ , we define  $Q_b(\varepsilon, k)$  as the following partial decision problem:*

- **Instance:** *An instance of MAX-E3-LIN-2 with  $m$  equations of the form  $x_{i_1} + x_{i_2} + x_{i_3} = b$ , each variable occurring exactly  $k$  times negated and  $k$  times unnegated.*
- **Objective:** *Decide if at least  $(1 + \varepsilon)m$  or at most  $(\frac{1}{2} - \varepsilon)m$  equations can be satisfied by an optimal assignment.*

To continue with the reduction, we will extend the NP-hardness results to a low occurrence version of  $w$ -MAX-E3-LIN-2, a weighted system of linear equations over  $\mathbb{Z}_2$  with either 2 or 3 variables. We call this a hybrid system of linear equations. Similarly, the task of the  $w$ -MAX-E3-LIN-2 problem is to find an assignment that maximizes the total weight of satisfied equations.

### 3 Reductions to $w$ -MAX-E3-LIN-2

#### 3.1 From $Q(\varepsilon, k)$ to $w$ -MAX-E3-LIN-2

Let  $\varepsilon \in (0, \frac{1}{4})$  and  $k > 0$  be an integer such that the problem  $Q(\varepsilon, k)$  is NP-hard. Let an instance  $I$  of  $Q(\varepsilon, k)$  be given by a set  $\nu(I)$  of variables, where  $\nu := |\nu(I)|$ . let's assume that there is an amplifier  $G = (V, E)$  with edge weights  $p: E \leftrightarrow \mathbb{R}^+$ , with a set of contacts  $D$  with  $|D| = k$ .

Now, there is a gap preserving reduction from  $Q(\varepsilon, k)$  to  $w$ -MAX-E3-LIN-2 with the amplifier as a parameter. The instance  $I$  of  $Q(\varepsilon, k)$  is transformed into a hybrid instance  $J$  of  $w$ -MAX-E3-LIN-2 as follows:

- For each variable  $x \in \nu(I)$ , we create a copy of the amplifier  $G$ , denoted  $G_x$ .
  - Inside  $G_x$ , the vertices correspond to variables in  $J$  and each edge  $(u, v)$  represents the equation  $u + v = 0$  with weight  $p(u, v)$ .
  - Contacts in  $G_x$  represent the  $k$  occurrences of the variable  $x$  in the equations of  $I$ . Each contact is associated with one occurrence of  $x$  in  $I$ .
- Every equation in  $I$  of the form  $x_{i_1} + x_{i_2} + x_{i_3} = b$  also belongs to  $J$  with weight 1.

Observe that the reduction above from  $I$  to  $J$  is a gap preserving reduction. Indeed, there is a dependency between the number of equations satisfied in  $I$  and the total weight of satisfied equations in  $J$ .

- If at least  $(1 + \varepsilon)m$  equations can be satisfied in  $I$ , then there is an assignment that satisfies all the equations in the amplifiers and at least  $(1 - \varepsilon)m$  equations of weight 1, leading to a total weight of satisfied equations of at least  $W_1 = k \cdot p(E) + (1 - \varepsilon)m$ .
- If at most  $(\frac{1}{2} - \varepsilon)m$  equations can be satisfied in  $I$ , then any assignment can satisfy at most  $\frac{1}{2}m - \varepsilon m$  equations of weight 1, leading to a total weight of satisfied equations of at most  $W_2 = k \cdot p(E) + (\frac{1}{2} - \varepsilon)m$ .

If we look at these problems as minimum unsatisfiability problems, where  $\text{OPT}'$  is the minimum weight of unsatisfied equations over all assignments, then  $\text{OPT}'(I) = \text{OPT}'(J)$ . Any assignment to variables from  $\nu(I)$  generate an assignment to variables of  $J$  naturally, as the value of  $x \in \nu(I)$  will be assigned to all the variables in  $G_x$ . These are called standard. Then,  $\text{OPT}'(J) \leq \text{OPT}'(I)$

because any assignment to  $I$  can be extended to a standard assignment to  $J$  with the same weight of unsatisfied equations.

To see that the optimum  $\text{OPT}'(J)$  is attained on standard assignments, we use the properties of the amplifier. Any assignment  $\varphi$  to the variables of  $J$  can be transformed into a standard assignment  $\varphi'$  without increasing the weight of unsatisfied equations. For this, we assign to all variables in  $G_x$  the value as it is assigned to the majority of contact vertices in  $G_x$  by  $\varphi$ . The fact that  $G$  is an amplifier guarantees (by definition) that the weight of unsatisfied equations in  $J$  does not increase in this transformation. After iterating this process for all variables, the result will be a standard assignment without increasing the weight of unsatisfied equations in  $J$ . Therefore,  $\text{OPT}'(J) = \text{OPT}'(I)$ .

### 3.2 From $Q_b(\varepsilon, 2k)$ to $w\text{-MAX-E3-LIN-2}$

The previous reduction is only the starting point, and a demonstration of how amplifiers can be useful. However, for the main results in this article, it is more convenient to start from the problem  $Q_b(\varepsilon, k)$ , where all equations have the same right hand side. This reduction is a slight modification of the previous one. Let  $\varepsilon \in (0, \frac{1}{4})$  and  $k > 0$  be an integer such that the problem  $Q_b(\varepsilon, k)$  is NP-hard. Assume that  $G = (V, E)$  is an amplifier with edge weights  $p : E \leftrightarrow \mathbb{R}^+$  and a set of contacts  $D$  with  $|D| = 2k$ . Let  $\{V^u, V^n\}$  be a partition of  $V$  balanced in  $D$ , so  $\|D \cap V^u\| = \|D \cap V^n\| = k$ . Also, let  $G^u$  and  $G^n$  be the induced subgraphs of  $G$  on  $V^u$  and  $V^n$ , respectively. Let an instance  $I$  of  $Q_b(\varepsilon, 2k)$  be given by a set  $\nu(I)$  of variables, where  $\nu := |\nu(I)|$ .

- For each variable  $x \in \nu(I)$ , we create a copy of the amplifier  $G$ , denoted  $G_x$ .
  - Any edge  $(u, u')$  in  $G_x^u$  or  $G_x^{u'}$  represents the equation  $u + u' = 0$  with weight  $p(u, u')$ . We call this cycle equations.
  - Any edge between  $v \in V_x^u$  and  $v' \in V_x^n$  represents the equation  $v + v' = 1$  with weight  $p(v, v')$ . We call this matching equations.
- The contact vertices of  $G_x^u$  (and of  $G_x^n$ ) represent the  $k$  occurrences of unnegated (respectively, negated) variable  $x$  in the equations of  $I$ . Every equation  $x + y + z = b$  from  $I$  also belongs to  $J$  with weight 1.

This way, we produce an instance  $J$  of  $w\text{-MAX-E3-LIN-2}$  with the same gap as in the previous reduction. As before, any assignment to variables from  $\nu(I)$  generate an assignment to variables of  $J$  naturally, as the value of  $x \in \nu(I)$  will be assigned to all the variables in  $G_x^u$ , and the value  $1 - x$  to all the variables in  $G_x^n$ . Once again, this assignment is called standard. The previous arguments applied to the problem  $Q(\varepsilon, 2k)$  can be translated to this new setting, by reversing all variable from  $V_x^n$  to their opposite.

### 3.3 Applying bi-wheel amplifiers

When we consider the existence of bi-wheel amplifiers  $(W_{k,\tau}, p)$  and add them to the framework we have just defined, we obtain the following theorem:

**Theorem 2** (Hybrid- $(W_{k,\tau}, p)$  inapproximability). *For every  $\varepsilon \in (0, \frac{1}{4})$ ,  $b \in 0, 1$  there exist instances of  $w\text{-MAX-E3-LIN-2}$ , which we call  $\text{hybrid-}(W_{k,\tau}, p)$ , with the following properties:*

- Each variable of the system of equations  $\text{hybrid-}(W_{k,\tau}, p)$  occurs exactly 3 times.
- $m$  equations are of the form  $x + y + z = b$  with weight 1.
- $3\tau m$  equations are of the form  $x + y = 0$ , with weight  $p_c$ .
- $\frac{3}{2}(\tau - 1)m$  equations are of the form  $x + y = 1$ , with weight  $p_m$ .

*It is NP-hard to decide if equations of weight at least  $(1 - \varepsilon)m$  or at most  $(\frac{1}{2} - \varepsilon)m$  can be satisfied by an optimal assignment.*

*Proof.* Let  $\varepsilon \in (0, \frac{1}{4})$ ,  $b \in 0, 1$  and  $k > 0$  be an integer such that the problem  $Q_b(\varepsilon, 2k)$  is NP-hard. Let an instance  $I$  of  $Q_b(\varepsilon, 2k)$  be given by a set  $\nu(I)$  of variables, where  $\nu := |\nu(I)|$ . Then  $m := |I| = \frac{2k\nu}{3}$ , as each variable occurs exactly  $2k$  times and each equation has 3 variables. For any fixed variable  $x \in \nu(I)$ , we create a copy of the bi-wheel amplifier  $W_{k,\tau}$ , denoted  $W_x$ . Each of these is a copy of  $W_{k,\tau}$ , with  $2\tau k$  variables  $\text{Var}(x) = \{x_j^u, x_{-j}^n\}_{j=1}^{\tau k}$ . Following the definition of a bi-wheel amplifier, each  $j^u$  is mapped to the variable  $x_j^u$  and  $j^n$  is mapped to  $x_j^n$ . We will denote the set of cycle edges as  $\mathcal{C}^u(W_x)$  and  $\mathcal{C}^n(W_x)$ , and the set of matching edges as  $\mathcal{M}(W_x)$ .

- For each matching edge  $(j^u, j'^n)$  in  $\mathcal{M}(W_x)$ , we create an equation  $x_j^u + x_{j'}^n = 1$  with weight  $p_m$ . There are  $\frac{3}{2}(\tau - 1)m$  such equations.
- For each cycle edge  $(j^q, (j + 1)^q)$  in  $\mathcal{C}^q(W_x)$ , we create an equation  $x_j^q + x_{j+1}^q = 0$  with weight  $p_c$ . We have  $2k\tau\nu = 3\tau m$  such equations.

Finally, we replace the  $l$ -th unnegated appearance of  $x$  in  $I$  with the variable  $x_{c^u(l)}^u := c_l^u$ , and the  $l$ -th negated appearance of  $x$  in  $I$  with the variable  $x_{c^n(l)}^n := c_l^n$ , for each  $l \in \{1, \dots, k\}$ . This doesn't add equations to the system, but it just renames the variables in the equations of  $I$  to the variables in the bi-wheel amplifiers.

Then, we have produced  $m$  equations of the form  $x + y + z = b$  with weight 1,  $3\tau m$  equations of the form  $x + y = 0$ , with weight  $p_c$ , and  $\frac{3}{2}(\tau - 1)m$  equations of the form  $x + y = 1$ , with weight  $p_m$ . The degree of the variables is 3 in any case, as each variable in the bi-wheel amplifier has degree 3, and each variable in the equations of  $I$  is replaced by a variable in the bi-wheel amplifier that also has degree 3. The gap preserving property of the reduction is a consequence of the properties of the bi-wheel amplifier, as mentioned in the previous section.  $\square$

It is at this point that we change the perspective, as this is the problem we will be using from now onwards. We will start by labeling all the variables in the system of equations  $\text{hybrid-}(W_{k,\tau}, p)$ . First of all, there are  $2\tau k\nu$  variables in the system, as each variable in  $\nu(I)$  is replaced by  $2\tau k$  variables in the bi-wheel amplifier. We label these variables as  $x_{i,j,b}$ , where  $i \in \{1, \dots, \nu\}$ ,  $j \in \{1, \dots, \tau k\}$  and  $b \in \{u, n\}$ . Each variable will participate in two distinct equations of the form  $x + y = 0$ , which correspond to the cycle equations. Additionally, each checker will appear in a single equation of the form  $x + y = 1$  and every contact will appear in a single equation of the form  $x + y + z = b$ . Another way of expressing this notion is that for  $S = \{\tau l : l \in \{1, \dots, k\}\}$ , and contacts are characterized by  $j \in S$ , while checkers are characterized by  $j \notin S$ . The weights of every equation are kept as in the definition of  $\text{hybrid-}(W_{k,\tau}, p)$ :  $p_c$  for cycle equations,  $p_m$  for matching equations and 1 for the equations of the form  $x + y + z = b$ .

## 4 Reduction to $\text{TSP}_f$ from $\text{Hybrid-}(W_{k,\tau}, p)$

This reduction is defined by gadgets that determine how we map every equation to a graph in which we can define a valid tour. We call these **equation gadgets**. We have seen that there are three types of equations in our system. The most interesting ones are the equations of the form  $x + y + z = b$ , and they define degree 3 equation gadgets.

**Definition 7** (Degree 3 equation gadget). *An **degree 3 equation gadget** is a weighted multigraph  $H = (V, E_u \dot{\cup} E_f, c)$  on a vertex set  $V = [4 + n_{aux}]$  with the following properties:*

- *The vertices are partitioned into three sets:*
  - *A central vertex, 4.*
  - *A set of three contact vertices,  $\{1, 2, 3\}$ .*
  - *A set of auxiliary vertices,  $\{5, 6, \dots, 4 + n_{aux}\}$ .*
- *$H$  forms an instance of  $\text{TSP}_f$ , and contains a fixed set of unforced edges  $E_s \subset E_u$ , called special edges, of cost  $1/2$  connecting the central vertex to the contacts.*

Given an equation gadget  $H$  and a valid tour  $Q$  in  $H$ , we define  $S_i^H(Q)$  as the number of special edges in  $Q$  that appear  $i$  times in  $Q$ . We aim to encode an assignment  $z \in \{0, 1\}^3$  to the variables in an equation as a spanning tour such that the special edge connecting the central vertex to the  $i$ -th contact appears twice if  $z_i = 1$  and does not appear if  $z_i = 0$ . The key idea is that satisfying assignments to the equation will correspond to valid tours with low weights and spanning tours not of this form will be more expensive. To encapsulate this, we define the following:

**Definition 8** (Soundness and completeness). *Given an equation gadget  $H = (V, E_u \dot{\cup} E_f, w)$  for an equation  $I = x + y + z = 1$ , we define:*

- **Completeness:** Let  $z \in \{0,1\}^3$ , and let  $C \subset \{0,1\}^3$  be the set of satisfying assignments of  $I$ . We use  $\mathcal{Q}_z$  to refer to the set of valid tours  $Q$  in  $H$  such that the special edge  $(\ell, 4)$  is unused for all  $\ell$  such that  $\mathbf{z}_\ell = 0$ , and is used twice otherwise.  $c(H)$  is defined as

$$c(H) = \max_{z \in C} \min_{Q \in \mathcal{Q}_z} c(Q)$$

- **Soundness:**  $s(H)$  is defined as

$$s(H) = \min_Q w(Q) - k_1^H(Q)/2 - \mathbb{1}_{k_1^H(Q)=0} \cdot \mathbb{1}_{k_2^H(Q) \text{ is even}}$$

where the minimum is taken over all valid spanning tours  $Q$ .

Now, we can provide a reduction from the NP-hard problem  $\text{hybrid-}(W_{k,\tau}, p)$  to  $\text{TSP}_f$ . Suppose we are given a  $\text{Hybrid-}(W_{k,\tau}, p)$  instance  $I$ , we will create an instance  $G$  of  $\text{TSP}_f$  as follows:

- We will create a single central vertex  $s$ , as well as a vertex  $v_{i,j,b}$  for each corresponding variable  $x_{i,j,b}$  in the system of equations.
- For each equation of the form  $x_{i,j,b} + x_{i',j',b'} = 0$ , we add to  $G$  a single unforced edge of cost  $p_c$  between  $v_{i,j,b}$  and  $v_{i',j',b'}$ .
- For each equation of the form  $x_{i,j,b} + x_{i',j',b'} = 1$ , we add to  $G$  two parallel forced edges of cost  $p_m$  between  $v_{i,j,b}$  and  $v_{i',j',b'}$ .
- For each equation of the form  $x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3} = 1$ , we add to  $G$  a copy of the gadget  $H$  corresponding to this equation, and we identify the contact vertices of  $H$  with the vertices  $v_{i_1,j_1,b_1}$ ,  $v_{i_2,j_2,b_2}$  and  $v_{i_3,j_3,b_3}$ , and the central vertex of  $H$  with  $s$ . The central vertex is shared between all such gadgets.

**Theorem 3.** Suppose there is an assignment  $x = (x_{i,j,b})_{i,j,b}$  to the  $\text{Hybrid-}(W_{k,\tau}, p)$  instance that satisfies all degree 2 clauses and violates at most  $\Delta$  degree 3 clauses. Then there is a spanning tour in  $G$  of total weight at most  $\text{papapa}$ .

*Proof.* We construct collections of edges  $T^=, T^/=$  AND  $T^3$  from  $G$  as follows:

- We add one copy of each forced edge of cost  $p_m$  to  $T^/=$ .
- If a variable  $x_{i,j,b}$  is set to 1,  $T^=$  contains one copy of both unforced edges of cost  $p_c$  corresponding to  $v_{i,j,b}$ .

The total weight of  $T^/=$  is  $2 \cdot (\tau - 1)p_mnk$ , while the total weight of  $T^=$  is  $\tau p_cnk$ . For each degree 3 clause  $x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3} = 1$ , we consider an optimal valid tour  $Q^* = \arg \min_{Q \in \mathcal{Q}_z} c(Q)$ . Then, we add the non-special edges of  $Q^*$  to  $T^3$ . Finally, we add to  $T^=$  two copies of the special edge  $(s, v_{i,j,b})$  for each variable  $x_{i,j,b}$  that is set to 0.  $\square$